PROVAL co.,Ltd

Calculation Sheet

Calculation header

Identifier	*New calculation (1)*

Medium selection and state

Medium	*Water*
State	*Liquid*

Operating data

☐ Safety-related application

	Maximum flow	Mean flow	Minimum flow	
t1	*60.0*	☒ *0.0*	☒ *0.0*	°C
p1	*5.0*	☒ *0.0*	☒ *0.0*	bar(a)
○ p2	*3.0*			bar(a)
◉ Δp	*2.0*	☒ *0.0*	☒ *0.0*	bar
○ qm	☒ *983,370.0*			kg/h
◉ qv	*1,000.0*	☒ *0.0*	☒ *0.0*	m^3/h
	Non-critical			

Fluid operating data

	Maximum flow	Mean flow	Minimum flow	
ϱ1	*983.37*	☒ *0.0*	☒ *0.0*	kg/m^3
ϱ2	*983.37*			kg/m^3
◉ η1	*0.466*	☒ *0.0*	☒ *0.0*	mPa s
○ v1	*0.47388*			mm^2/s
pv1	*0.19946*	☒	☒	bar(a)
tv1	*151.83*	*373.95*	*373.95*	°C
cF1	*1,551.7*			m/s

Pipe downstream of valve

Size class downstream of valve	NPS2	*2"*
Schedule downstream	SCH2	*STD*

Valve configuration

Valve type		*Straight globe valve*	
Trim type		*Contoured plug*	
Flow direction		*FTC*	
Valve performance class		*Multi stage valve*	
Protection		*Non-hardened*	
Low-noise design		*n.a.*	

Valve data

Basic characteristic		*Linear*	
☐ Lift stopper			
Selected valve size	NPS ☒	*2"*	
Pressure class	class	*class 600*	
Maximum operating pressure	p,max	*105.0*	bar(a)
Valve modifier (Cv/Cv100 = 0.75)	F_L^2	*0.917*	-
Valve modifier (Cv/Cv100 = 0.75)	Kc	*0.877*	-
Cavitation modifier (Cv/Cv100 = 0.75)	xFz	*0.604*	-
Valve style modifier (Cv/Cv100 = 1)	Fd	*0.41*	-

Load-dependent values

	Maximum flow	Mean flow	Minimum flow	
u2	*137.05*			m/s

Noise calculation

Calculation standard (noise, liquid)	*IEC 60534-8-4 (2015)*

Noise prediction data

Low-noise design		*n.a.*	
Sound velocity	cF1	*1,551.7*	m/s
Pressure level	LA1	*165.0*	dB
Sound pressure level	Lpi	*165.0*	dB

Legend

Errors